Week 6: Product Review and Rating Management System

Scenario:

The e-commerce enterprise seeks to collect shopper impressions regarding fulfilled orders. Verified customers are invited to post numeric ratings and qualitative feedback once their purchases arrive. The database layer must securely store user testimonials, evaluate product quality metrics, and distinguish high-performing merchandise across catalog categories.

Students will design and implement a Product Review and Rating Module integrated with the existing Customer and Product master records.

Tasks to be Done:

1. Create Review and Rating Tables

```
SQL> CREATE TABLE PRODUCT_REVIEWS(
  2  FEEDBACK_ID VARCHAR2(7),
  3  CUST_REF VARCHAR2(7),
  4  PROD_REF VARCHAR2(7),
  5  RATING_SCORE NUMBER(1),
  6  USER_FEEDBACK VARCHAR2(50),
  7  POSTED_DATE DATE,
  8  PRIMARY KEY(FEEDBACK_ID),
  9  FOREIGN KEY(CUST_REF) REFERENCES Customer(Customer_ID),
 10  FOREIGN KEY(PROD_REF) REFERENCES Product(Product_ID));

Table created.

SQL> desc PRODUCT_REVIEWS
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 FEEDBACK_ID                               NOT NULL VARCHAR2(7)
 CUST_REF                                           VARCHAR2(7)
 PROD_REF                                           VARCHAR2(7)
 RATING_SCORE                                       NUMBER(1)
 USER_FEEDBACK                                      VARCHAR2(50)
 POSTED_DATE                                        DATE
```

Store Customer Feedback and Ratings

Tasks:

1. Add reviews for purchased products.

```
SQL> INSERT INTO PRODUCT_REVIEWS
  2  (FEEDBACK_ID, CUST_REF, PROD_REF, RATING_SCORE, USER_FEEDBACK, POSTED_DATE)
  3  VALUES
  4  ('REV0001', 'CUST000001', 'P000001', 5, 'Excellent laptop performance', DATE '2026-01-20');

1 row created.

SQL>
SQL> INSERT INTO PRODUCT_REVIEWS
  2  (FEEDBACK_ID, CUST_REF, PROD_REF, RATING_SCORE, USER_FEEDBACK, POSTED_DATE)
  3  VALUES
  4  ('REV0002', 'CUST000001', 'P000002', 4, 'Good wireless mouse', DATE '2026-01-21');

1 row created.

SQL>
SQL> INSERT INTO PRODUCT_REVIEWS
  2  (FEEDBACK_ID, CUST_REF, PROD_REF, RATING_SCORE, USER_FEEDBACK, POSTED_DATE)
  3  VALUES
  4  ('REV0003', 'CUST000002', 'P000004', 5, 'Very good sound quality', DATE '2026-02-02');

1 row created.

SQL>
SQL> INSERT INTO PRODUCT_REVIEWS
  2  (FEEDBACK_ID, CUST_REF, PROD_REF, RATING_SCORE, USER_FEEDBACK, POSTED_DATE)
  3  VALUES
  4  ('REV0004', 'CUST000002', 'P000006', 4, 'Good quality USB cable', DATE '2026-02-03');

1 row created.

SQL>
SQL> INSERT INTO PRODUCT_REVIEWS
  2  (FEEDBACK_ID, CUST_REF, PROD_REF, RATING_SCORE, USER_FEEDBACK, POSTED_DATE)
  3  VALUES
  4  ('REV0005', 'CUST000003', 'P000005', 5, 'Excellent smartphone', DATE '2026-02-15');

1 row created.
```

```
SQL> SELECT * FROM PRODUCT_REVIEWS;

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0001 CUST000001 P000001               5
Excellent laptop performance                       20-JAN-26

REV0002 CUST000001 P000002               4
Good wireless mouse                                21-JAN-26

REV0003 CUST000002 P000004               5
Very good sound quality                            02-FEB-26

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0004 CUST000002 P000006               4
Good quality USB cable                             03-FEB-26

REV0005 CUST000003 P000005               5
Excellent smartphone                              15-FEB-26

REV0006 CUST000004 P000008               4
Useful smart watch                                 28-FEB-26

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0007 CUST000004 P000006               5
Cable works perfectly                              01-MAR-26

REV0008 CUST000005 P000010               4
Good tablet for daily use                          12-MAR-26

REV0009 CUST000006 P000009               5
Clear and comfortable headphones                   25-MAR-26

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0010 CUST000006 P000015               4
Good sound and battery life                        26-MAR-26

REV0011 CUST000007 P000008               5
Excellent features                                 10-APR-26

REV0012 CUST000008 P000012               4
Reliable storage device                            30-APR-26

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0013 CUST000008 P000006               3
Average quality                                    01-MAY-26

REV0014 CUST000010 P000012               5
Very useful external drive                         28-MAY-26

REV0015 CUST000011 P000009               4
Good headphones                                    15-JUN-26

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0016 CUST000011 P000015               5
Excellent earbuds                                  16-JUN-26

REV0017 CUST000012 P000014               4
Useful laptop stand                                01-JUL-26

REV0018 CUST000012 P000003               5
Excellent keyboard                                 02-JUL-26

18 rows selected.
```

2. Store customer ratings from 1 to 5 stars.

Ensure this by adding Constraint

```
SQL> ALTER TABLE PRODUCT_REVIEWS ADD CONSTRAINT CHK_RATING_RANGE CHECK(RATING_SCORE >= 1 AND RATING_SCORE <=

Table altered.

SQL> INSERT INTO PRODUCT_REVIEWS
  2  VALUES ('REV0019', 'CUST000001', 'P000001', 6,
  3          'Excellent product', SYSDATE);
INSERT INTO PRODUCT_REVIEWS
*
ERROR at line 1:
ORA-02290: check constraint (C##ECOMMERCE.CHK_RATING_RANGE) violated
```

3. Update review comments if required.

```
SQL> UPDATE PRODUCT_REVIEWS
  2  SET USER_FEEDBACK = 'Good quality but could be improved'
  3  WHERE FEEDBACK_ID = 'REV0013';

1 row updated.
```

4. Remove inappropriate or invalid reviews.

```
SQL> DELETE FROM PRODUCT_REVIEWS
  2  WHERE FEEDBACK_ID = 'REV0018';

1 row deleted.
```

4. Retrieve Product Review Details

Write SQL queries to:

1. Display all reviews for a product.

```
SQL> SELECT * FROM PRODUCT_REVIEWS WHERE PROD_REF = 'P000012';

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0012 CUST000008 P000012                 4
Reliable storage device                            30-APR-26

REV0014 CUST000010 P000012                 5
Very useful external drive                         28-MAY-26
```

2. Display customer name with their reviews.

```
SQL> SELECT CONCAT(c.First_Name, c.Last_Name) AS Customer_Name,
  2  R.FEEDBACK_ID, R.PROD_REF, R.RATING_SCORE, R.USER_FEEDBACK
  3  FROM Customer c JOIN PRODUCT_REVIEWS R
  4  ON c.Customer_ID = R.CUST_REF;

CUSTOMER_NAME                  FEEDBAC PROD_RE    RATING_SCORE
------------------------------ ------- ------- ---------------
USER_FEEDBACK
--------------------------------------------------
ArjunKumar                     REV0001 P000001               5
Excellent laptop performance

ArjunKumar                     REV0002 P000002               4
Good wireless mouse

PriyaSharma                    REV0003 P000004               5
Very good sound quality

CUSTOMER_NAME                  FEEDBAC PROD_RE    RATING_SCORE
------------------------------ ------- ------- ---------------
USER_FEEDBACK
--------------------------------------------------
PriyaSharma                    REV0004 P000006               4
Good quality USB cable

RahulMenon                     REV0005 P000005               5
Excellent smartphone

AnanyaIyer                     REV0006 P000008               4
Useful smart watch

CUSTOMER_NAME                  FEEDBAC PROD_RE    RATING_SCORE
------------------------------ ------- ------- ---------------
USER_FEEDBACK
--------------------------------------------------
AnanyaIyer                     REV0007 P000006               5
Cable works perfectly

VikramReddy                    REV0008 P000010               4
Good tablet for daily use

MeeraKrishnan                  REV0009 P000009               5
Clear and comfortable headphones

CUSTOMER_NAME                  FEEDBAC PROD_RE    RATING_SCORE
------------------------------ ------- ------- ---------------
USER_FEEDBACK
--------------------------------------------------
MeeraKrishnan                  REV0010 P000015               4
Good sound and battery life

SureshNair                     REV0011 P000008               5
Excellent features

KavyaRajan                     REV0012 P000012               4
Reliable storage device
```

```
CUSTOMER_NAME                  FEEDBAC PROD_RE    RATING_SCORE
------------------------------ ------- ------- ---------------
USER_FEEDBACK
--------------------------------------------------
KavyaRajan                     REV0013 P000006               3
Good quality but could be improved

DivyaPillai                    REV0014 P000012               5
Very useful external drive

AdityaVerma                    REV0015 P000009               4
Good headphones

CUSTOMER_NAME                  FEEDBAC PROD_RE    RATING_SCORE
------------------------------ ------- ------- ---------------
USER_FEEDBACK
--------------------------------------------------
AdityaVerma                    REV0016 P000015               5
Excellent earbuds

SnehaPatel                     REV0017 P000014               4
Useful laptop stand

17 rows selected.
```

3. Find products having maximum reviews.

```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       COUNT(R.FEEDBACK_ID) AS REVIEW_COUNT
  5  FROM PRODUCT P
  6  JOIN PRODUCT_REVIEWS R
  7    ON P.PRODUCT_ID = R.PROD_REF
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY REVIEW_COUNT DESC;

PRODUCT PRODUCT_NAME                                    REVIEW_COUNT
------- ---------------------------------------------- ------------
P000006 USB Cable                                                 3
P000008 Smart Watch                                               2
P000012 External Hard Disk                                        2
P000015 Wireless Earbuds                                          2
P000009 Headphones                                                2
P000014 Laptop Stand                                              1
P000010 Tablet                                                    1
P000005 Smartphone                                                1
P000001 Laptop                                                    1
P000004 Bluetooth Speaker                                         1
P000002 Wireless Mouse                                            1

11 rows selected.
```

4. Display recent customer feedback.

```
SQL> SELECT
  2       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3       P.PRODUCT_NAME,
  4       R.RATING_SCORE,
  5       R.USER_FEEDBACK AS CUSTOMER_FEEDBACK,
  6       R.POSTED_DATE AS REVIEW_DATE
  7  FROM PRODUCT_REVIEWS R
  8  JOIN CUSTOMER C
  9    ON R.CUST_REF = C.CUSTOMER_ID
 10  JOIN PRODUCT P
 11    ON R.PROD_REF = P.PRODUCT_ID
 12  ORDER BY R.POSTED_DATE DESC
 13  FETCH FIRST 5 ROWS ONLY;

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME                                       RATING_SCORE
-------------------------------------------------- ----------------
CUSTOMER_FEEDBACK                                  REVIEW_DA
-------------------------------------------------- ---------
Sneha Patel
Laptop Stand                                                4
Useful laptop stand                                01-JUL-26

Aditya Verma
Wireless Earbuds                                            5
Excellent earbuds                                  16-JUN-26

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME                                       RATING_SCORE
-------------------------------------------------- ----------------
CUSTOMER_FEEDBACK                                  REVIEW_DA
-------------------------------------------------- ---------
Aditya Verma
Headphones                                                  4
Good headphones                                    15-JUN-26

Divya Pillai
External Hard Disk                                          5
Very useful external drive                         28-MAY-26

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME                                       RATING_SCORE
-------------------------------------------------- ----------------
CUSTOMER_FEEDBACK                                  REVIEW_DA
-------------------------------------------------- ---------
Kavya Rajan
USB Cable                                                   3
Good quality but could be improved                 01-MAY-26
```

5. Retrieve reviews with ratings above 4.

```
SQL> SELECT * FROM PRODUCT_REVIEWS WHERE RATING_SCORE > 4;

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0001 CUST000001 P000001               5
Excellent laptop performance                       20-JAN-26

REV0003 CUST000002 P000004               5
Very good sound quality                            02-FEB-26

REV0005 CUST000003 P000005               5
Excellent smartphone                              15-FEB-26

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0007 CUST000004 P000006               5
Cable works perfectly                              01-MAR-26

REV0009 CUST000006 P000009               5
Clear and comfortable headphones                   25-MAR-26

REV0011 CUST000007 P000008               5
Excellent features                                 10-APR-26

FEEDBAC CUST_RE PROD_RE    RATING_SCORE
------- ------- ------- ---------------
USER_FEEDBACK                                      POSTED_DA
-------------------------------------------------- ---------
REV0014 CUST000010 P000012               5
Very useful external drive                         28-MAY-26

REV0016 CUST000011 P000015               5
Excellent earbuds                                  16-JUN-26

8 rows selected.
```

Example Query:

5. Calculate Average Product Ratings

Use aggregate functions to analyze ratings.

Tasks:

1. Calculate average rating for each product.

```
SQL> SELECT p.Product_Name, AVG(R.RATING_SCORE) AS Average_Rating
  2  FROM Product p JOIN PRODUCT_REVIEWS R
  3  ON p.Product_ID = R.PROD_REF
  4  GROUP BY p.Product_Name;

PRODUCT_NAME                                       AVERAGE_RATING
-------------------------------------------------- --------------
Laptop                                                          5
Wireless Mouse                                                  4
Bluetooth Speaker                                               5
Smartphone                                                      5
USB Cable                                                       4
Smart Watch                                                   4.5
Headphones                                                    4.5
Tablet                                                          4
External Hard Disk                                            4.5
Laptop Stand                                                    4
Wireless Earbuds                                              4.5

11 rows selected.
```

2. Count the number of reviews per product.

```
SQL> SELECT p.Product_Name, Count(R.RATING_SCORE) AS Count_Rating
  2  FROM Product p JOIN PRODUCT_REVIEWS R
  3  ON p.Product_ID = R.PROD_REF
  4  GROUP BY p.Product_Name;

PRODUCT_NAME                                       COUNT_RATING
-------------------------------------------------- --------------
Laptop                                                        1
Wireless Mouse                                                1
Bluetooth Speaker                                             1
Smartphone                                                    1
USB Cable                                                     3
Smart Watch                                                   2
Headphones                                                    2
Tablet                                                        1
External Hard Disk                                            2
Laptop Stand                                                  1
Wireless Earbuds                                              2

11 rows selected.
```

3. Find highest-rated products.

```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       ROUND(AVG(R.RATING_SCORE), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  JOIN PRODUCT_REVIEWS R
  7    ON P.PRODUCT_ID = R.PROD_REF
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING AVG(R.RATING_SCORE) = (
 10       SELECT MAX(AVG_RATING)
 11       FROM (
 12            SELECT AVG(RATING_SCORE) AS AVG_RATING
 13            FROM PRODUCT_REVIEWS
 14            GROUP BY PROD_REF
 15       )
 16  );

PRODUCT PRODUCT_NAME                                  AVERAGE_RATING
------- -------------------------------------------------- --------------
P000001 Laptop                                                     5
P000004 Bluetooth Speaker                                          5
P000005 Smartphone                                                 5
```

4. Identify products with average rating above 4.

```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       ROUND(AVG(R.RATING_SCORE), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  JOIN PRODUCT_REVIEWS R
  7    ON P.PRODUCT_ID = R.PROD_REF
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING AVG(R.RATING_SCORE) > 4
 10  ORDER BY AVERAGE_RATING DESC;

PRODUCT PRODUCT_NAME                                  AVERAGE_RATING
------- -------------------------------------------------- --------------
P000005 Smartphone                                                 5
P000004 Bluetooth Speaker                                          5
P000001 Laptop                                                     5
P000012 External Hard Disk                                       4.5
P000009 Headphones                                               4.5
P000015 Wireless Earbuds                                         4.5
P000008 Smart Watch                                              4.5

7 rows selected.
```

6. Identify Highly Rated Products

Generate product rating reports.

Report 1: Product Rating Analysis

Display:

- Product name
- Number of reviews
- Average rating

```
SQL> SELECT
  2       P.PRODUCT_NAME,
  3       COUNT(R.FEEDBACK_ID) AS NUMBER_OF_REVIEWS,
  4       ROUND(AVG(R.RATING_SCORE), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  LEFT JOIN PRODUCT_REVIEWS R
  7    ON P.PRODUCT_ID = R.PROD_REF
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY AVERAGE_RATING DESC NULLS LAST;

PRODUCT_NAME                                       NUMBER_OF_REVIEWS
-------------------------------------------------- -----------------
AVERAGE_RATING
--------------
Bluetooth Speaker                                                  1
             5

Smartphone                                                         1
             5

Laptop                                                             1
             5

Smart Watch                                                        2
           4.5

Wireless Earbuds                                                   2
           4.5

External Hard Disk                                                 2
           4.5

Headphones                                                         2
           4.5

USB Cable                                                          3
             4

Wireless Mouse                                                     1
             4

Laptop Stand                                                       1
             4

Tablet                                                             1
             4

Mechanical Keyboard                                                0


Web Camera                                                         0


Gaming Monitor                                                     0


Power Bank                                                         0


15 rows selected.
```

Report 2: Customer Feedback Analysis

Find:

- Most reviewed products

```
SQL> SELECT
  2       P.PRODUCT_NAME,
  3       COUNT(R.FEEDBACK_ID) AS REVIEW_COUNT
  4  FROM PRODUCT P
  5  JOIN PRODUCT_REVIEWS R
  6    ON P.PRODUCT_ID = R.PROD_REF
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY REVIEW_COUNT DESC;

PRODUCT_NAME                                       REVIEW_COUNT
-------------------------------------------------- ------------
USB Cable                                                     3
Smart Watch                                                   2
External Hard Disk                                            2
Wireless Earbuds                                              2
Headphones                                                    2
Laptop Stand                                                  1
Tablet                                                        1
Smartphone                                                    1
Laptop                                                        1
Bluetooth Speaker                                             1
Wireless Mouse                                                1

11 rows selected.
```

- Highly rated products

```
SQL> SELECT
  2       P.PRODUCT_NAME,
  3       ROUND(AVG(R.RATING_SCORE), 2) AS AVERAGE_RATING
  4  FROM PRODUCT P
  5  JOIN PRODUCT_REVIEWS R
  6    ON P.PRODUCT_ID = R.PROD_REF
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  HAVING AVG(R.RATING_SCORE) > 4
  9  ORDER BY AVERAGE_RATING DESC;

PRODUCT_NAME                                       AVERAGE_RATING
-------------------------------------------------- --------------
Smartphone                                                      5
Bluetooth Speaker                                               5
Laptop                                                          5
External Hard Disk                                            4.5
Headphones                                                    4.5
Wireless Earbuds                                              4.5
Smart Watch                                                   4.5

7 rows selected.
```

- Products requiring improvement

```
SQL> SELECT
  2       P.PRODUCT_NAME,
  3       COUNT(R.FEEDBACK_ID) AS REVIEW_COUNT,
  4       ROUND(AVG(R.RATING_SCORE), 2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  JOIN PRODUCT_REVIEWS R
  7    ON P.PRODUCT_ID = R.PROD_REF
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING AVG(R.RATING_SCORE) < 3
 10  ORDER BY AVERAGE_RATING;

no rows selected
```

Report 3: Rating Distribution

- Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS FIVE_STAR_RATINGS
  2  FROM PRODUCT_REVIEWS
  3  WHERE RATING_SCORE = 5;

FIVE_STAR_RATINGS
-----------------
                8
```

- Number of 4-star ratings

```
SQL> SELECT COUNT(*) AS FOUR_STAR_RATINGS
  2  FROM PRODUCT_REVIEWS
  3  WHERE RATING_SCORE = 4;

FOUR_STAR_RATINGS
-----------------
                8
```

- Number of low-rated products

```
SQL> SELECT COUNT(*) AS LOW_RATED_PRODUCTS
  2  FROM (
  3       SELECT PROD_REF
  4       FROM PRODUCT_REVIEWS
  5       GROUP BY PROD_REF
  6       HAVING AVG(RATING_SCORE) < 3
  7  );

LOW_RATED_PRODUCTS
------------------
                 0
```

- Complete Rating

```
SQL> SELECT
  2       RATING_SCORE,
  3       COUNT(*) AS NUMBER_OF_RATINGS
  4  FROM PRODUCT_REVIEWS
  5  GROUP BY RATING_SCORE
  6  ORDER BY RATING_SCORE DESC;

RATING_SCORE NUMBER_OF_RATINGS
------------ -----------------
           5                 8
           4                 8
           3                 1
```